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A Multimodal artificial intelligence framework for mechanistic discovery and digital twin construction in cardiovascular medicine [☆]



To the Editor

Over the past decade, AI models that integrate information from multiple clinical sources, including electrocardiograms, cardiac imaging, laboratory tests, electronic health records, and wearable devices, have achieved remarkable performance in automated detection, granular classification, and outcome prediction across a wide spectrum of cardiovascular conditions (Fig. 1). These successes have demonstrated the unique strength of AI in processing multidimensional and heterogeneous medical data and have positioned it as a powerful assistant in routine cardiology practice.^{1,2} From ECG analysis for identifying complex arrhythmias to precise quantification of plaque burden in coronary CT, AI has become a powerful tool to improve clinical diagnostic efficiency. But if AI is still limited to incremental benefits such as accelerated diagnosis or improved risk prediction, its value will be greatly reduced. We stand at the threshold of a new era where multimodal AI must evolve from being a “powerful diagnostic tool” to becoming a “catalyst” driving fundamental scientific discoveries and a “simulator” empowering forward-looking personalized therapies. The next frontier now focuses on two transformative directions: decoding the black-box mechanisms of diseases and building dynamic digital twins of patients. The former seeks to answer “why diseases occur,” while the latter aims to explore “how to develop optimal treatment plans for specific individuals.”

1. From Correlation to Causality: Mechanistic Discovery Enabled by Multimodal AI

Genome-wide association studies (GWAS) have uncovered hundreds of genetic loci associated with complex cardiovascular traits. However, most of these signals stay at the level of population-scale statistical association and do not immediately tell us which organ, cell population, spatial niche, or regulatory circuit actually executes the risk conferred by a given variant.³ This gap frequently leaves clinicians and biologists in a “we know that it is associated, but we do not know how it causes disease” dilemma. The power of multimodal AI lies in its ability to build a “bridge” from macroscopic image phenotypes to microscopic molecular activities, enabling closed-loop interpretation from genes to functions.

1.1. Integrating multi-omics to infer causal networks

Recent deep learning architectures, including Transformers, multimodal encoders with attention mechanisms, and graph neural networks, can ingest measurements derived from cardiac MRI (e.g. left ventricular ejection fraction, strain, myocardial fibrosis burden), coronary CT plaque characteristics, and temporally resolved flow features together with genomics, transcriptomics, proteomics, and metabolomics. A landmark example is the DeepFlow-like framework applied to phase-contrast cardiac MRI data from the UK Biobank.⁴ By automatically segmenting the aorta and left ventricle, the model generated high-throughput quantitative phenotypes such as forward aortic flow, stroke volume, and regurgitant fraction. GWAS performed on physiologically meaningful traits derived by AI identified novel loci that would have been missed with conventional, manually derived imaging measurements. Among these, a variant in the PLN gene, which encodes phospholamban, a central regulator of cardiomyocyte calcium cycling, was associated with stroke volume, delineating a mechanistic chain from genotype to calcium handling perturbation to impaired contractility and ultimately organ level pump dysfunction.⁵ This is more than correlation; it is a data-driven reconstruction of a plausible causal pathway linking molecular perturbation to organ-level phenotype.

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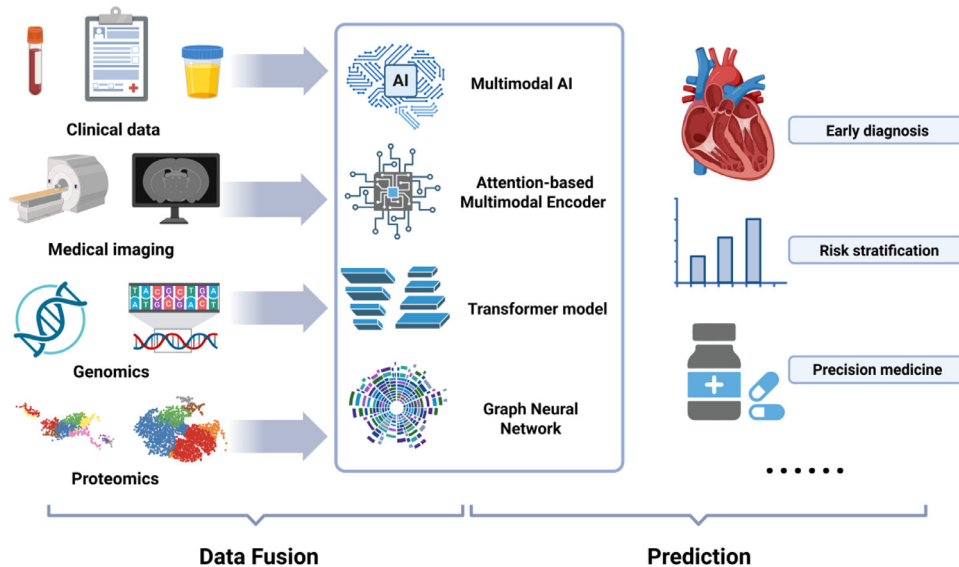


Fig. 1. AI models capable of synthesizing data from diverse clinical origins.

1.2. Discovering new targets and mechanistic subtypes

This deep multimodal integration capability opens up revolutionary ways for drug discovery and patient stratification. In abdominal aortic aneurysm (AAA), Zhang and colleagues built an AI-driven framework that coupled clinical variables, radiomics signatures, and multi-omics profiles within the same patient cohort.⁶ Their model resolved at least two biologically distinct subtypes. One subtype was characterized by strong infiltration of macrophages, T cells, and neutrophils and by broad activation of inflammatory pathways such as NF- κ B, IL-1 β signaling, chemotaxis, and antigen presentation. The other subtype was dominated by extracellular matrix (ECM) turnover, matrix metalloproteinase-mediated collagen and elastin degradation, and disordered stromal metabolism. This mechanistic stratification explained why aneurysms with similar diameters can progress at different rates and, more importantly, suggested that the subtype driven by immunity and inflammation may benefit more from antiinflammatory or immunomodulatory therapies, whereas the ECM remodeling subtype might require interventions targeting protease activity or promoting vascular wall repair. Idiomyopathy, integrating whole-exome sequencing with cardiac MRI has similarly shown that patients with different monogenic backgrounds present with distinct patterns of myocardial remodeling, fibrosis, and ventricular outflow obstruction, which in turn inform prognosis and surveillance strategies.⁷ Together, these studies illustrate a larger trend in cardiovascular medicine: we are transitioning from symptom-based and image-based classification to mechanism-anchored, AI-refined subtyping, in which the subtype definition already embeds a therapeutic hypothesis.

2. From Static to Dynamic: Cardiovascular Digital Twins Powered by Multimodal AI

A digital twin, in the cardiovascular context, refers to a dynamic, high-fidelity, computational replica of an individual patient's heart and vascular system. Such a twin must encode precise anatomy, electrophysiology, and hemodynamics, and it must be updatable as new data arrive. Achieving this level of personalization is extremely challenging if we rely only on traditional, labor-intensive image processing pipelines. Multimodal AI provides the missing automation and the ability to fuse multiple data sources into a consistent, simulation-ready model (Fig. 2).

2.1. Create 4D dynamic heart

By jointly processing high-resolution coronary CT for anatomy, cine cardiac MRI for myocardial motion, T1/T2 mapping or late-gadolinium enhancement for tissue characterization, and echocardiography for real-time dynamics, AI models can generate a patient-specific 4D heart model (three-dimensional space plus time). Kong and Shadden proposed a deep learning method for whole-heart mesh generation that learns to deform a template heart mesh to match the patient's CT/MR images.⁸ The proposed method enables the direct generation of a topologically consistent, high-quality volumetric mesh suitable for computational fluid dynamics and finite-element simulation, thereby entirely circumventing the conventional multi-step workflow involving segmentation, surface reconstruction, mesh cleaning, and remeshing. This approach significantly reduces the barrier to large-scale digital twin generation.

2.2. Integrating into the therapeutic workflow

This dynamic digital twin technology revolutionizes clinical decision support. Before undergoing interventional treatments for structural heart diseases like transjugular aortic valve replacement (TAVR), physicians can utilize a patient-specific digital twin to simulate implantation outcomes of various valve models and sizes in a virtual environment. Through personalized computational models based on the patient's digital twin, doctors can pre-simulate different valve configurations, release depths, and sizes in virtual scenarios. This enables preoperative evaluation of perivalvular leakage (PVL) risks, potential coronary artery compression or occlusion, and post-implantation hemodynamic changes in the left ventricular outflow tract and ascending aorta. The system achieves “first-time optimal” personalized surgical planning, significantly reducing complication risk.⁹ Beyond surgical interventions, digital twins offer transformative potential in chronic disease management. By integrating real-time physiological data from wearable devices like smartwatches and implantable cardiac monitors, these digital twins function as dynamic models that continuously update to reflect the patient's evolving cardiac status.¹⁰ On this basis, doctors can simulate the progression of diseases (such as heart failure) and predict the long-term results of different drug treatment strategies or lifestyle interventions, providing dynamic and prospective decision support for patients.

3. Implementation path and core challenges

Despite the promising prospects, there are still many severe challenges to achieve these two major frontier goals, and it is urgent to tackle them through interdisciplinary collaboration.

3.1. Data and algorithm bottlenecks

The realization of deep mechanism discovery and the construction of high-fidelity digital twins rely on large-scale, rigorously annotated, and standardized multimodal longitudinal datasets. At the algorithmic level, research must be directed toward developing next-generation fusion models capable of precisely capturing complex nonlinear cross-modal correlations while maintaining inherent interpretability. The ability to articulate a clear decision-making rationale is crucial for fostering clinical trust. Additionally, high-resolution, multi-physical-field digital twin simulations impose substantial computational demands, posing a significant challenge to their practical deployment. Optimizing algorithms to achieve near-real-time simulations in clinical scenarios remains a key technical challenge requiring urgent breakthroughs.

3.2. Clinical translation and validation

To validate their effectiveness and clinical utility, these advanced AI applications must undergo rigorously designed prospective multicenter randomized controlled trials. The core of this validation is to determine whether they can genuinely improve patient

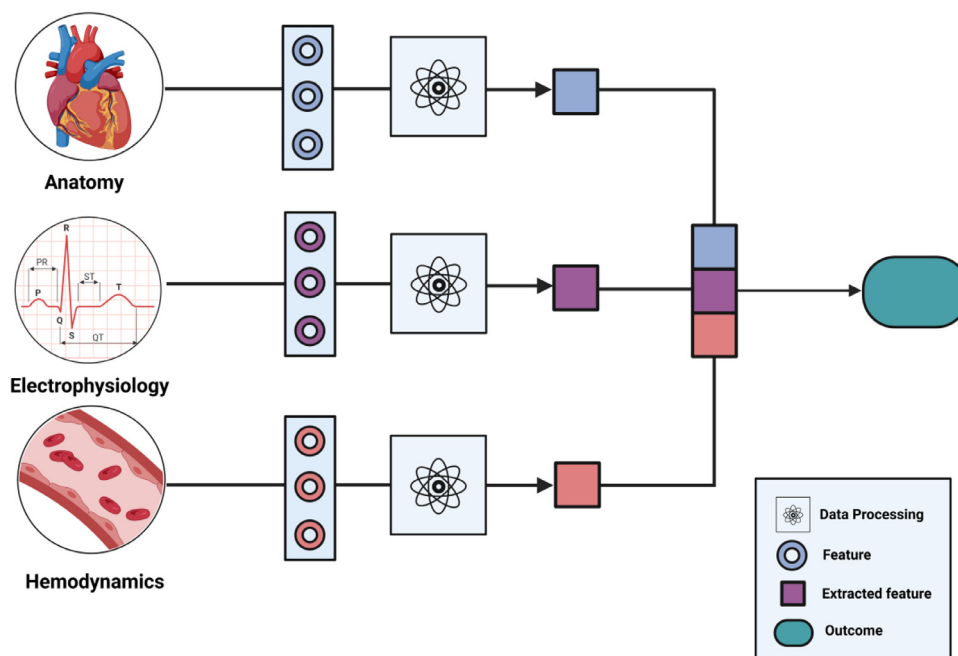


Fig. 2. Multimodal AI Workflow for Dynamic Cardiovascular Digital Twins.

outcomes. The key challenge lies in transforming the complex multidimensional information generated by these models (such as genetic risk scores and simulated blood flow velocity maps) into decision support tools that clinicians can easily understand and implement, which is crucial for transitioning from “technical prototypes” to “clinical standard practice”.

3.3. Ethics, privacy, and regulation

Digital twins involve extremely sensitive physiologic replicas of individual patients. Their creation, storage, sharing, and continuous updating mandate new paradigms for data ownership, patient consent, cybersecurity, and lifecycle governance. Regulatory agencies such as the U.S. FDA or China's NMPA will need agile but rigorous frameworks for approving AI-powered, self-learning, and continuously evolving software-as-a-medical-device products, striking a balance between innovation and risk control.

4. Conclusion

The journey of multimodal AI in cardiovascular medicine is transitioning from successful tool applications to a more exciting chapter of scientific empowerment. Moving beyond mere “accurate observation,” it now aims to achieve “thorough understanding” and “precise prediction.” By decoding the intrinsic molecular mechanisms of diseases and dynamically simulating individual cardiac physiology and pathology, multimodal AI holds the potential to fundamentally accelerate cardiovascular drug development cycles, reshaping clinical paradigms rooted in simulation and prediction. This future cannot be achieved by any single discipline alone, but urgently necessitates a concerted effort in deep integration and collaborative innovation across clinical medicine, molecular biology, computational science, data ethics, and government regulation. Only through such synergy can we collectively propel cardiovascular medicine beyond current boundaries, ushering in a new era of true precision, personalization, and predictive capabilities.

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GZ, JX and GZ designed and designed the research; GZ, JX, ZZ, CY and GZ drew the diagram; GZ, JX, ZZ, CY, KK and GZ wrote the manuscript, DL, MD and GZ reviewed wrote the manuscript. All authors read and approved the final manuscript.

Ethics Approval and Consent to Participate

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Availability of Data and Materials

Not applicable.

Disclosure of Interest

None.

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